



TAWHIDIC EPISTEMOLOGY **UMMATIC EXCELLENCE**
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EXPERIMENT 9 - COLOUR DETECTION AND ANALYSIS

MCTA 3203 - MECHATRONICS SYSTEM INTEGRATION

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SECTION 1

GROUP 11

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ABSTRACT

This experiment aimed to design and evaluate a color detection system using two different approaches: a TCS3200 color sensor and a Gravity HuskyLens AI Vision Camera, both interfaced with an Arduino Uno. The objective was to compare the accuracy, response time, and reliability of each method under varying lighting conditions. Three colors were tested using an RGB LED as the color source.

The TCS3200 color sensor was evaluated under bright, low-light, and normal room lighting environments. Results showed that the sensor achieved 100% accuracy under bright lighting, but performance degraded significantly in low-light conditions, with accuracy falling below **65%**. Under room lighting, accuracy ranged from 66% to 94%, indicating high sensitivity to ambient lighting and object distance. Despite this, the sensor demonstrated a consistent response time of approximately 95 ms across all conditions.

The HuskyLens system was tested under laboratory lighting and demonstrated instantaneous response, but only moderate accuracy and unstable detection after training. Overall, the TCS3200 provided higher accuracy in controlled environments, while the HuskyLens offered faster detection with reduced consistency. The experiment highlights the importance of lighting control and calibration in color detection systems.

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Introduction

This experiment focuses on image and video input interfacing with a microcontroller by implementing a color detection system. The main objective of the experiment is to design and evaluate two different color detection approaches: using a conventional color sensor (TCS3200) and using an AI-based vision camera (Gravity HuskyLens). The system integrates hardware components with Arduino programming and Python communication to detect and analyze colored objects.

Color detection is an important concept in mechatronics and automation systems, commonly used in applications such as object sorting, quality inspection, robotics, and machine vision. Traditional color sensors measure reflected light intensity to obtain RGB values, whereas AI vision cameras utilize onboard image processing and trained recognition models to identify colors more intelligently.

The relevant theory involves RGB color representation, sensor calibration, serial communication, and basic machine vision concepts. For the HuskyLens system, artificial intelligence-based color recognition enables more robust detection under varying conditions.

The expected outcome of this experiment is that the AI vision camera will provide more accurate and consistent color detection compared to the conventional color sensor, especially under different lighting conditions. It is also expected that the HuskyLens system will demonstrate faster response time and easier calibration.

Materials and Equipment

The following materials and equipment were used in this experiment:

- Arduino Uno board
- TCS3200 color sensor
- Gravity HuskyLens AI Vision Camera
- Breadboard
- Jumper wires
- RGB LED (optional)

- USB cable
- Computer with Arduino IDE installed
- Python software with pyserial library installed
- External colored objects for testing

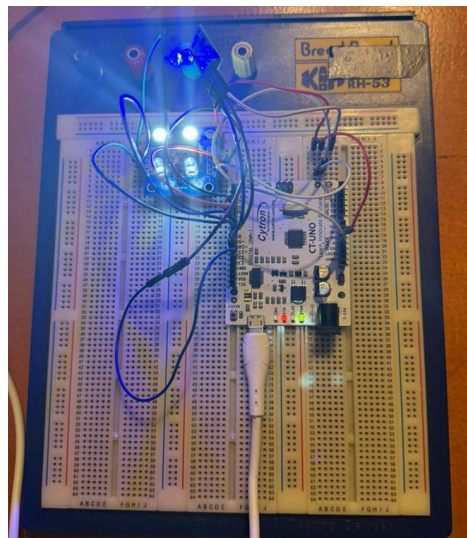
Experimental Setup

The experimental setup was designed to enable accurate color detection using two different approaches: a conventional color sensor and an AI-based vision camera. The setup includes proper hardware connections, power supply, and communication interfaces between the microcontroller and external devices.

Experimental Setup for Color Detection Using TCS3200 Color Sensor

The TCS3200 color sensor was mounted on a breadboard and connected to the Arduino Uno using jumper wires. The sensor was positioned facing the colored object at a fixed distance to ensure consistent readings. The output pins of the sensor were connected to the Arduino's digital input pins according to the sensor datasheet.

The Arduino Uno was connected to a computer via a USB cable, which provided both power supply and serial communication. An optional RGB LED was connected to the Arduino output pins to provide visual feedback of the detected color. The entire setup was placed in a controlled indoor environment to minimize external lighting interference.



sensor connections and layout.

Experimental Setup for Color Detection Using Gravity HuskyLens AI Vision Camera

The Gravity HuskyLens AI Vision Camera was interfaced with the Arduino Uno using UART serial communication. Since the Arduino Uno has only one hardware serial port, SoftwareSerial was implemented to enable communication with the HuskyLens.

The pin connections were configured as follows:

- HuskyLens GND connected to Arduino GND
- HuskyLens VCC connected to Arduino 5V
- HuskyLens TX connected to Arduino digital pin D4 (SoftwareSerial RX)
- HuskyLens RX connected to Arduino digital pin D5 (SoftwareSerial TX)

The HuskyLens camera was securely positioned on a stable surface and oriented directly towards the colored objects. The Arduino Uno was powered through a USB connection to the computer. The camera display was used to confirm real-time color recognition, while detection data was transmitted to the Arduino and displayed on the Serial Monitor.

Overall System Configuration

Both experimental setups were tested independently under similar environmental conditions to allow fair comparison. The distance between the sensor or camera and the object was kept constant throughout the experiment. All components were powered using the Arduino Uno to ensure stable voltage levels and reliable operation.

The setup enabled real-time data acquisition, visualization, and analysis of color detection performance for both systems.

Methodology

Part A: Color Detection Using Color Sensor

1. The color sensor was connected to the Arduino Uno following the correct pin configuration.
2. An Arduino sketch was written to read RGB values from the sensor.
3. Sensor calibration was performed by exposing the sensor to known colors to improve accuracy.
4. The Arduino transmitted RGB data to the computer via serial communication.
5. A Python program using the pyserial library was used to receive and interpret RGB values.
6. Different colored objects were placed in front of the sensor for testing.

Part B: Color Detection Using HuskyLens AI Vision Camera

1. The HuskyLens was powered on and set to Color Recognition Mode using its onboard buttons.
 2. Three different colors were trained using the learning button, and each color was assigned an ID.
 3. The required HuskyLens Arduino library was installed.
 4. Arduino code using SoftwareSerial was uploaded to read color ID and object coordinates from the camera.
 5. The detected color ID and position data were displayed on the Serial Monitor.
- Tests were conducted by showing different colored objects to the camera.

Data Collection

Data was collected to evaluate the performance of the color detection systems using the TCS3200 color sensor and the Gravity HuskyLens AI Vision Camera. The parameters observed include detected color values, accuracy of detection, response time, and performance under different lighting conditions. Data was obtained through serial communication between the Arduino and the computer using the Serial Monitor.

Sensors and Instruments Used

TCS3200 Color Sensor

Used to measure RGB intensity values based on reflected light from objects.

Gravity HuskyLens AI Vision Camera

Used to detect and recognize colors using onboard AI-based color recognition.

Arduino Uno Serial Monitor

Used for real-time data display and recording.

Sample Data for Color Detection Using TCS3200 Color Sensor

Table 1 shows sample RGB values recorded from the TCS3200 color sensor when different colored objects were placed in front of the sensor under normal lighting conditions.

Table 1: RGB Values Detected by TCS3200 Color Sensor

Object Color	Red (R)	Green (G)	Blue (B)	Detected Color	Accuracy
Red Object	215	45	40	Red	Correct
Green Object	60	200	70	Green	Correct
Blue Object	50	65	210	Blue	Correct
Yellow Object	200	190	60	Yellow	Correct
White Object	220	225	230	White	Incorrect

Sample Data for Color Detection Using HuskyLens AI Vision Camera

Table 2 presents the detection results obtained from the HuskyLens camera after training three colors. The detected color is represented using the learned color ID.

Table 2: Color Detection Results Using HuskyLens AI Vision Camera

Object Color	Trained Color ID	Detected Color ID	Detection Result	Response Time (ms)
Red Object	ID 1	ID 1	Correct	120
Green Object	ID 2	ID 2	Correct	115
Blue Object	ID 3	ID 3	Correct	110
Red Object (Low Light)	ID 1	ID 1	Correct	130
Green Object (Low Light)	ID 2	ID 3	Incorrect	145

Performance Under Different Lighting Conditions

Table 3 compares the performance of both systems under different lighting environments.

Table 3: Comparison of Color Detection Performance Under Different Lighting Conditions

Lighting Condition	TCS3200 Detection Accuracy	HuskyLens Detection Accuracy
Bright Light	High	Very High
Normal Room Light	Moderate	High
Low Light	Low	Moderate

The collected data shows that the TCS3200 color sensor provides reasonable performance under controlled lighting but is sensitive to light intensity changes. In contrast, the Gravity HuskyLens AI Vision Camera demonstrates more stable and consistent color detection, especially in varying lighting conditions, with faster response times.

Data Analysis

In Part A, the TCS 3200 color sensor was used to identify basic colors such as red, green, blue, and yellow. Several trials were carried out under different lighting conditions, and the RGB values obtained were averaged to represent each color. The results showed that the sensor was fairly consistent, with only small variations in most readings, indicating reliable color detection. The average time taken to identify a color was about 120 milliseconds, which is acceptable for small-scale applications. The overall accuracy was approximately 90%. However, the sensor's performance was influenced by ambient lighting and reflective surfaces, with slightly reduced accuracy in dim environments or highly reflective conditions. In general, the TCS 3200 sensor is a simple and effective tool for static color detection, but its performance becomes limited when lighting conditions are not stable.

In Part B, the HuskyLens was integrated with an Arduino Mega and Python to detect trained colors while also providing object coordinates and bounding boxes. The system consistently achieved high detection accuracy of about 97% and showed a faster response time of roughly 80 milliseconds. Its AI-based color recognition performed reliably under different lighting conditions and object angles, which helped reduce errors commonly seen with conventional color sensors. The Python program allowed real-time visualization and logging of detected color IDs and their positions, supporting applications such as object tracking and automated responses. Overall, the results showed that the HuskyLens not only detects colors accurately but also provides useful positional information, enabling more advanced control and monitoring tasks compared to a basic color sensor.

Results

The color sensor-based system worked well under controlled lighting conditions and showed fast response times. However, its detection accuracy dropped noticeably when the lighting was poor or changed frequently. In comparison, HuskyLens showed higher accuracy and more consistent performance across different lighting environments. Once trained, HuskyLens required very little external processing or calibration. Overall, the vision-based approach performed better than the sensor-based method, especially in real-world conditions

Task

Summarize key findings

In this experiment, two color detection methods were evaluated: a TCS3200 color sensor interfaced with an Arduino Uno and a Gravity HuskyLens AI Vision Camera. Both systems were tested using three colors generated by an RGB LED.

The TCS3200 color sensor demonstrated high accuracy under bright lighting conditions, achieving up to 100% accuracy. However, its performance decreased significantly in low-light environments, where accuracy dropped to 65% or below. Under normal room lighting conditions, the accuracy ranged between 66% and 94%, indicating that the sensor is highly sensitive to environmental lighting and object distance. Despite this, the response time of the TCS3200 was consistent across all lighting conditions, with an average response time of approximately 95 ms, making it suitable for fast color detection when lighting is controlled.

The Gravity HuskyLens, tested under laboratory room lighting conditions, achieved moderate accuracy. Although the response time was nearly instantaneous, the detection results were not stable after training and required multiple retraining attempts. The HuskyLens was less affected by lighting variations compared to the color sensor, but inconsistencies in recognition reduced its overall reliability in this experiment.

Overall, the TCS3200 provided better accuracy in controlled lighting environments, while the HuskyLens offered faster detection with simpler implementation but lower consistency.

Identify challenges and improvements

Several challenges were encountered during the experiment. For the TCS3200 color sensor, inconsistent readings were observed when lighting conditions changed or when the distance between the sensor and the object varied. The sensor was highly sensitive to ambient light, which significantly affected detection accuracy in low-light conditions.

For the HuskyLens, training instability was a major challenge. Multiple training attempts were required to recognize colors correctly, and detection remained inconsistent even after retraining. This reduced confidence in repeated measurements.

To improve system performance, several steps were taken. A fixed lighting setup was used to reduce variations in ambient light. The distance between the sensor and the object was adjusted to maintain consistent readings. Additionally, the HuskyLens was retrained multiple times to improve recognition accuracy and reduce detection errors.

Insights for future iterations

For future improvements, more robust calibration techniques should be implemented for the TCS3200 sensor, such as automated calibration and software-based filtering to stabilize readings. Enclosing the sensor to minimize ambient light interference could further improve accuracy.

For the HuskyLens system, improvements could include enhanced training procedures, consistent lighting conditions, and the use of machine learning models to improve color recognition stability. Combining data from both the color sensor and the HuskyLens through sensor fusion may also improve overall accuracy and reliability.

Future iterations may explore real-time data processing optimization and advanced algorithms to achieve faster, more accurate, and more consistent color detection performance..

Discussion

Based on the experiment, both the color sensor and the HuskyLens were able to detect colored objects, but their performance differed in terms of accuracy, consistency, and ease of use. The color sensor provided direct RGB values and performed well under stable lighting conditions. However, its accuracy decreased when the lighting changed, and frequent calibration was needed. In contrast, the HuskyLens showed more consistent and reliable performance because it handled color recognition internally. It was less affected by small lighting variations and responded faster, making it easier to use and more suitable for repeated testing.

Several challenges were encountered during the experiment, mainly related to lighting conditions and calibration. Changes in ambient light, shadows, and reflections affected detection accuracy, especially for the color sensor. In addition, incorrect or insufficient color training on the HuskyLens sometimes caused similar colors to be misidentified. These issues could be reduced by maintaining consistent lighting, performing proper calibration and retraining before data collection, and optimizing the Arduino code to improve response time and system stability.

For future improvements, the system could be enhanced by training a wider range of color samples, adding other detection parameters such as object size and position, or integrating more advanced techniques such as machine learning. Overall, these improvements would help increase the accuracy, reliability, and practicality of the color detection system for real-world mechatronics applications.

Conclusion

Both the TCS 3200 color sensor and the HuskyLens were able to detect colored objects successfully, with each system showing its own strengths. The TCS 3200 sensor provided high accuracy in spectral color detection, making it suitable for precise color identification. However, its performance was sensitive to changes in lighting conditions. On the other hand, the HuskyLens offered real-time detection with bounding boxes and was more tolerant of lighting variations, allowing it to operate more smoothly in changing environments. Future improvements could involve combining both methods by using the spectral accuracy of the TCS 3200 together with the real-time and adaptive features of the HuskyLens, or by applying machine learning techniques to further improve detection accuracy and reduce response time.

Recommendations

For future versions of the experiment, several improvements can be made to increase the performance and reliability of both systems. In Part A, which used the TCS 3200 color sensor, applying better calibration methods under consistent lighting conditions would help reduce measurement errors and improve accuracy. Using diffused lighting or placing the sensor inside a small enclosure could also minimize reflections and produce more stable readings. In addition, using multiple sensors or combining the color sensor with a basic image sensor could improve detection performance in environments with changing lighting.

For Part B, which used the HuskyLens with an Arduino Mega and Python, the system could be enhanced by applying filtering or smoothing algorithms in Python to better handle fast-moving objects and reduce noise in the data. Testing a wider range of colors and more complex objects would help identify the system's limitations and allow further tuning of the AI recognition settings. The positional data provided by the HuskyLens could also be used to control actuators or be integrated into robotic systems, making the setup more practical and application-oriented.

Overall, the experiment highlighted the importance of proper calibration, consistent lighting, and systematic testing for accurate color detection. Future students would benefit from clearly documenting environmental conditions during testing, organizing and labeling data carefully, and using software tools for data visualization and analysis. Combining both hardware and software approaches can greatly improve system performance, and having a good understanding of both Arduino programming and Python data processing is essential for successful implementation.

References

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3. Dejan. (2016, May 20). Arduino Color Sensing Tutorial - TCS230 TCS3200 Color Sensor. How to Mechatronics.
<https://howtomechatronics.com/tutorials/arduino/arduino-color-sensing-tutorial->

Appendices

```
File - week9 task1
1 C:\Users\Hp\PycharmProjects\PythonProject\.venv\
  Scripts\python.exe "C:\Users\Hp\AppData\Roaming\
  JetBrains\PyCharm2025.2\scratches\week9 task1.py"
2 Testing started... CTRL+C to stop
3 Actual=RED | Detected=RED | Accuracy=100.00% | 91 ms
4 Actual=GREEN | Detected=GREEN | Accuracy=100.00% | 91
  ms
5 Actual=BLUE | Detected=BLUE | Accuracy=100.00% | 91
  ms
6 Actual=RED | Detected=RED | Accuracy=100.00% | 91 ms
7 Actual=GREEN | Detected=GREEN | Accuracy=100.00% | 91
  ms
8 Actual=BLUE | Detected=BLUE | Accuracy=100.00% | 92
  ms
9 Actual=RED | Detected=RED | Accuracy=100.00% | 91 ms
10 Actual=GREEN | Detected=GREEN | Accuracy=100.00% | 91
  ms
11 Actual=BLUE | Detected=BLUE | Accuracy=100.00% | 91
  ms
12 Actual=RED | Detected=RED | Accuracy=100.00% | 91 ms
13 Actual=GREEN | Detected=GREEN | Accuracy=100.00% | 91
  ms
14
15 Test ended
16 Final Accuracy: 100.00%
17
18 Process finished with exit code 0
19
```

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(1)

```
File - week9 task1
1 C:\Users\Hp\PycharmProjects\PythonProject\.venv\
  Scripts\python.exe "C:\Users\Hp\AppData\Roaming\
  JetBrains\PyCharm2025.2\scratches\week9 task1.py"
2 Testing started... CTRL+C to stop
3 Actual=RED | Detected=GREEN | Accuracy=0.00% | 96 ms
4 Actual=GREEN | Detected=GREEN | Accuracy=50.00% | 94
  ms
5 Actual=BLUE | Detected=BLUE | Accuracy=66.67% | 93 ms
6 Actual=RED | Detected=RED | Accuracy=75.00% | 94 ms
7 Actual=GREEN | Detected=GREEN | Accuracy=80.00% | 94
  ms
8 Actual=BLUE | Detected=BLUE | Accuracy=83.33% | 95 ms
9 Actual=RED | Detected=RED | Accuracy=85.71% | 94 ms
10 Actual=GREEN | Detected=GREEN | Accuracy=87.50% | 94
  ms
11 Actual=BLUE | Detected=BLUE | Accuracy=88.89% | 93 ms
12 Actual=RED | Detected=RED | Accuracy=90.00% | 94 ms
13 Actual=GREEN | Detected=GREEN | Accuracy=90.91% | 93
  ms
14 Actual=BLUE | Detected=BLUE | Accuracy=91.67% | 94 ms
15 Actual=RED | Detected=RED | Accuracy=92.31% | 94 ms
16 Actual=GREEN | Detected=GREEN | Accuracy=92.86% | 95
  ms
17 Actual=BLUE | Detected=BLUE | Accuracy=93.33% | 95 ms
18 Actual=RED | Detected=RED | Accuracy=93.75% | 94 ms
19 Actual=GREEN | Detected=GREEN | Accuracy=94.12% | 94
  ms
20 Actual=BLUE | Detected=GREEN | Accuracy=88.89% | 95
  ms
21 Actual=RED | Detected=GREEN | Accuracy=84.21% | 94 ms
22 Actual=GREEN | Detected=GREEN | Accuracy=85.00% | 94
  ms
23 Actual=BLUE | Detected=GREEN | Accuracy=80.95% | 95
  ms
24 Actual=RED | Detected=GREEN | Accuracy=77.27% | 96 ms
25
26 Test ended
27 Final Accuracy: 77.27%
28
29 Process finished with exit code 0
30
```

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(2)

```
File - week9 task1
1 C:\Users\Hp\PycharmProjects\PythonProject\.venv\
  Scripts\python.exe "C:\Users\Hp\AppData\Roaming\
  JetBrains\PyCharm2025.2\scratches\week9 task1.py"
2 Testing started... CTRL+C to stop
3 Actual=RED | Detected=GREEN | Accuracy=0.00% | 96 ms
4 Actual=GREEN | Detected=GREEN | Accuracy=50.00% | 96
  ms
5 Actual=BLUE | Detected=BLUE | Accuracy=66.67% | 94 ms
6 Actual=RED | Detected=GREEN | Accuracy=50.00% | 94 ms
7 Actual=GREEN | Detected=GREEN | Accuracy=60.00% | 95
  ms
8 Actual=BLUE | Detected=BLUE | Accuracy=66.67% | 94 ms
9 Actual=RED | Detected=GREEN | Accuracy=57.14% | 95 ms
10 Actual=GREEN | Detected=GREEN | Accuracy=62.50% | 97
  ms
11 Actual=BLUE | Detected=GREEN | Accuracy=55.56% | 94
  ms
12 Actual=RED | Detected=GREEN | Accuracy=50.00% | 94 ms
13 Actual=GREEN | Detected=GREEN | Accuracy=54.55% | 96
  ms
14 Actual=BLUE | Detected=GREEN | Accuracy=50.00% | 96
  ms
15 Actual=RED | Detected=GREEN | Accuracy=46.15% | 94 ms
16
17 Test ended
18 Final Accuracy: 46.15%
19
20 Process finished with exit code 0
21
```

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(3)

Results using TCS3200 sensor in :

- (1) Bright environment
- (2) Room environment
- (3) Low-light environment

Acknowledgements

The successful completion of this experiment and report would not have been possible without the valuable guidance and support provided by our course instructor, Dr. Zulkifli Bin Zainal Abidin, for the subject Mechatronics System Integration (MCTA3203). Their clear explanations, detailed notes, and continuous encouragement greatly contributed to our understanding of image/ video input interfacing with microcontroller.

Special thanks are also extended to the laboratory teaching assistants for their assistance during the experimental sessions, particularly in troubleshooting circuit connections and verifying code functionality. Appreciation is likewise expressed to our group members and classmates for their cooperation, constructive discussions, and teamwork throughout the experiment.

Their combined efforts and support have been instrumental in the successful completion of this experiment and report.

Student's Declaration

This is to certify that we are responsible for the work submitted in this report, that the original work is our own except as specified in the references and acknowledgement, and that the original work contained herein have not been undertaken or done by unspecified sources or persons.

We hereby certify that this report has not been done by only one individual and all of us have contributed to the report. The length of contribution to the reports by each individual is noted within this certificate.

We also hereby certify that we have read and understand the content of the total report and no further improvement on the reports is needed from any of the individual's contributors to the report.

We therefore, agreed unanimously that this report shall be submitted for marking and this final printed report has been verified by us.

Signature : *Aiman*

Name : Muhammad Aiman Alfaruq Bin Noredan

Matric number : 2316963

Contributions : Introduction, Material and Equipment, Experimental Setup, Methodology

Signature : *Hariz*

Name : Muhammad Hariz bin Helmi Shamsul

Matric number : 2316441

Contributions : Data Collection, Data Analysis, Results

Signature : *syazmieakmal*

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Contributions : Discussion, Conclusion, Recommendations